

Gaël Röthlin

Full-Stack & IoT Developer – Master 1 IoT Student

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About Me

Student in the [International Master 1 IoT](#) ([EIPHI Graduate School](#), [Université de Marie & Louis Pasteur](#)) after a Bachelor's degree (BUT) in Computer Science, I design systems from sensor to cloud: distributed architectures, full-stack applications, and local, low-tech AI infrastructure. I am looking for a work-study placement (alternance) starting September 2026 (2026–2027) in IoT, embedded systems, or full-stack development.

Professional Experience

3rd-Year BUT Internship, IoT & Edge Architecture Engineer

Jan. – Apr. 2026

Animation Numérique Innovante & Low-Tech

Dijon, France

- Designed an edge / on-premise infrastructure for local, low-tech AI.
- Developed a 3D anatomical object visualization tool in virtual reality to support teaching for professors and researchers.
- Led projects at the Crunch Time event (UTBM hackathon, Belfort-Montbéliard).

2nd-Year BUT Internship, Full-Stack Developer, Euphron Project

Apr. – May 2025

Learning Centre Claude Oytana

Besançon, France

- Web/mobile application facilitating the donation of sports equipment between students.
- Development across the entire application stack, front-end and back-end.

Education

International Master 1 IoT (Internet of Things)

Since Sept. 2026

[EIPHI Graduate School](#) – [Université de Marie & Louis Pasteur](#), Montbéliard (25)

Data Science & AI · Embedded Systems · Advanced Networks · Mobile Dev · Programmable Matter · Smart Cities · Cloud & Virtualization

BUT Computer Science – 7th/47 (3rd year)

2023 – 2026

[IUT Nord Franche-Comté](#), [Université de Marie & Louis Pasteur](#), Belfort (90)

Full-stack development · Deployment · Virtualization

Technological Baccalaureate STI2D – Mention Bien (High Honors)

2021 – 2023

[SIN Track](#), [Lycée Édouard Belin](#), Belfort (90)

Embedded Systems · Electronics · 3D Modeling · Machining

Projects

Pluto | [KiCad](#), [ZMK](#) ([Zephyr](#)), [BLE](#), [Nice!Nano v2](#) ([nRF52840](#))

Mar. – Jul. 2026

- Designed and built a split ergonomic Bluetooth keyboard from scratch: schematic and layout ([ergogen.xyz](#)), PCB routing ([KiCad](#), manufactured via [JLCPCB](#)), SMD soldering (diodes, hotswap sockets, [Nice!Nano](#) controller), [ZMK](#) firmware with multi-OS Bluetooth profile management, chassis designed and 3D-printed (PLA/PETG).

Sagittarius | [Vue 3](#), [Ollama](#)

Dec. 2025 – Feb. 2026

- Multi-workspace local AI web platform (Code, Mail, Calendar, Notes) connected to [Ollama](#), with no cloud dependency.

Smart-Waste | [Java](#), [ESP32](#), [Docker](#), [Cucumber](#)/[JUnit](#)

Sept. 2025 – Jun. 2026

- Distributed IoT system for smart urban waste management: multi-threaded TCP server driving [ESP32](#) sensors, MongoDB storage with REST API and degraded-mode fallback, waste type recognition via image analysis.

SpotCLI | [C](#), [POSIX Sockets](#), [Spotify Web API](#)

Oct. 2025 – Mar. 2026

- Command-line Spotify client built in C from scratch: 38-command TUI, OAuth 2.0 authentication with a custom HTTP callback mini-server, full wrapper of the Spotify API.

Skills

Internet of Things (IoT) & Embedded : Arduino, Raspberry Pi, C, Lua

Mobile : Kotlin, Swift, React Native, Quasar

Web & Backend Development : VueJS, NodeJS, PHP, Python, Java

Systems, GUI & Scripting : C++, Qt, Bash, AWK, LaTeX

DevOps & Infrastructure : Linux, Git, Docker, Kubernetes, Proxmox

Other : PSC1 First Aid Certification (Sept. 2026) · Player, volunteer, and activity leader at the Cité Internationale des Jeux (CIJ) – since Sept. 2026.